

A Literature-Implied Complex Subcase of Brech–Piña’s Dual-Isometry Question

Research note for run `fa_banach_001`

18 July 2026

Status. Literature-implied answer (partial subcase). The countable complex-scalar version of Brech–Piña’s Question 6 has a positive answer by Fakhoury’s later theorem on isometries of combinatorial-like Banach spaces.

Source question. Brech and Piña define, for a compact hereditary family $\mathcal{F} \subseteq [\kappa]^{<\omega}$ containing the singletons, the combinatorial space $X_{\mathcal{F}}$ as the completion of $c_{00}(\kappa)$ for

$$\|x\|_{\mathcal{F}} = \sup_{s \in \mathcal{F}} \sum_{\alpha \in s} |x(\alpha)|.$$

After proving their primal Banach–Stone theorem under the hypothesis $[\kappa]^1 \subseteq \overline{\mathcal{F}^{MAX}} \cap \overline{\mathcal{G}^{MAX}}$, they ask:

Is every isometry $T : X_{\mathcal{F}}^* \rightarrow X_{\mathcal{G}}^*$ weak*-weak* continuous?

Equivalently, is every such T the adjoint of an isometry $S : X_{\mathcal{G}} \rightarrow X_{\mathcal{F}}$? This is Question 6 in Section 2 of [1].

Supporting theorem. Fakhoury works with countable combinatorial families on $\mathbb{N}$ and with the more general spaces $X_{\mathcal{F}, \mathbf{E}}$. In Corollary 2.15, specialized to the complex scalar field and $\mathbf{E} = \ell_1$, she obtains that if $\mathcal{F}_1, \mathcal{F}_2$ are compact combinatorial families, then every isometry

$$J : X_{\mathcal{F}_1}^* \longrightarrow X_{\mathcal{F}_2}^*$$

is an $(\mathcal{F}_1, \mathcal{F}_2)$ -compatible signed permutation of the coordinate functionals. This is also stated in Proposition 1.3(b) for one family: if $\mathbb{K} = \mathbb{C}$ and $\mathcal{F}$ is compact, every isometry of $X_{\mathcal{F}}^*$ is an $\mathcal{F}$ -compatible signed permutation. See [2, Proposition 1.3 and Corollary 2.15].

Implication. A compatible signed permutation on the dual basis has the form

$$J e_i^* = \omega_i e_{\pi(i)}^*, \quad |\omega_i| = 1, \quad \pi(\mathcal{F}_1) = \mathcal{F}_2.$$

The corresponding operator on the primal spaces sends $e_{\pi(i)}$ to $\omega_i e_i$ (with the evident inverse convention) and is an isometry because π transports the family. Therefore J is the adjoint of a primal isometry. In particular it is weak*-weak* continuous. Thus Brech–Piña’s Question 6 is answered positively when $\kappa = \mathbb{N}$ and the scalar field is $\mathbb{C}$, even without the extra assumption that every singleton lies in the closure of the maximal sets.

Scope and limitations. This packet does not settle the real scalar case. Fakhoury explicitly notes that the analogous arbitrary-family real statement is false, citing work in progress of Brech–Tcaciuc, but that warning does not by itself decide the real case under Brech–Piña’s stronger closure-of-maximals hypothesis. The packet also does not address uncountable κ .

Search evidence. The target was checked against the run indexes for 2007.11071, **Banach–Stone**, **combinatorial Banach**, **weak***, and **dual isometry**. No prior packet for this exact paper was found. Web and local-source searches for the source title, the displayed weak*-continuity question, and “ $X_{\mathcal{F}}^*$ combinatorial Banach isometry” led to Fakhoury’s arXiv:2305.13125. Later papers on dual quasi-norms and the Zoo of combinatorial Banach spaces did not appear to answer the weak*-continuity question.

References

- [1] C. Brech and C. Piña, *Banach–Stone-like results for combinatorial Banach spaces*, Ann. Pure Appl. Logic 172 (2021), no. 8, Paper No. 102989; arXiv:2007.11071.
- [2] M. Fakhoury, *Isometries of p -convexified combinatorial Banach spaces*, Studia Mathematica (2024); arXiv:2305.13125.